

Exam Summary

Module Name

Student

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Week 1: Linear Algebra for Deep Learning

Chapter 1: Linear Algebra for Deep Learning

Part 1 — The Building Blocks

Type	Dims	Notation	Example
Scalar	0D	a	single loss value
Vector	1D	$\mathbf{x}$ — always column	$[1, 2, 3]^\top$
Matrix	2D	$\mathbf{A}$, element $A_{i,j}$	weight layer
Tensor	3D+	$\mathbf{A}$, element $A_{i,j,k}$	image batch

Column vector and matrix notation:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \mathbf{A} = \begin{pmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{pmatrix}$$

Part 2 — Operations

Matrix addition (same size only):

- Used for: combining two sets of features, adding residuals in ResNets

$$C_{i,j} = A_{i,j} + B_{i,j}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 10 & 12 \end{pmatrix}$$

Scalar on matrix (a = scale, c = bias/shift):

- Used for: normalising data, applying a weight + bias to a whole layer

$$D_{i,j} = a \cdot B_{i,j} + c$$

$$2 \cdot \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + 10 = \begin{pmatrix} 12 & 14 \\ 16 & 18 \end{pmatrix} \rightarrow \text{every element } \times 2, \text{ then } +10$$

Broadcasting (vector added to every column of matrix):

- Used for: adding a bias vector to an entire batch of inputs at once

$$C_{i,j} = A_{i,j} + b_j$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} + \begin{pmatrix} 10 \\ 20 \end{pmatrix} = \begin{pmatrix} 11 & 22 \\ 13 & 24 \\ 15 & 26 \end{pmatrix}$$

Matrix multiplication:

- *Used for:* the core forward pass — transforming inputs through a weight layer: $\mathbf{y} = W\mathbf{x}$

$$\mathbf{C} = \mathbf{AB} \quad C_{i,j} = \sum_k A_{i,k} \cdot B_{k,j}$$

Shape: $(m \times k) \cdot (k \times n) = (m \times n)$ — inner dims must match.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

Inner / dot product (vectors only, result is a scalar):

- *Used for:* measuring similarity between two vectors; attention scores in Transformers

$$\mathbf{x}^\top \mathbf{y} = \sum_i x_i y_i = \|\mathbf{x}\|_2 \|\mathbf{y}\|_2 \cos \theta$$

$$(1 \ 2 \ 3) \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 = 32$$

Hadamard (element-wise) product:

- *Used for:* gating mechanisms (LSTM, GRU), applying masks, dropout

$$\mathbf{C} = \mathbf{A} \odot \mathbf{B} \quad C_{i,j} = A_{i,j} \cdot B_{i,j}$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \odot \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 5 & 12 \\ 21 & 32 \end{pmatrix}$$

Properties of matrix multiplication:

- Distributive: $A(B + C) = AB + AC$
- Associative: $A(BC) = (AB)C$

Exam Trap: $AB \neq BA$ — matrix multiplication is **NOT commutative**. This is on every exam.

Transpose:

- *Used for:* converting column↔row vectors, computing dot products ($\mathbf{x}^\top \mathbf{y}$), flipping weight matrices in backpropagation

$$(A^\top)_{i,j} = A_{j,i} \quad (AB)^\top = B^\top A^\top$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^\top = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \rightarrow (2 \text{ times } 3) \text{ becomes } (3 \text{ times } 2)$$

Part 3 — Norms

- *Used for:* measuring magnitude of weights, gradients, errors; regularisation (L1/L2); comparing vectors

General L^p norm:

$$\|x\|_p = \left(\sum_i |x_i|^p \right)^{\frac{1}{p}}, \quad p \geq 1$$

Example with $x = [3, -4]^\top$:

$$\|x\|_1 = 3 + 4 = 7 \quad \|x\|_2 = \sqrt{9 + 16} = 5 \quad \|x\|_\infty = 4$$

Norm	Formula	Use
L^2 (Euclidean)	$\ x\ _2 = \sqrt{\sum_i x_i^2}$	most common
Squared L^2	$\ x\ _2^2 = x^\top x$	easier to compute
L^1	$\ x\ _1 = \sum_i x_i $	sparsity
L^0	count of non-zeros	not a true norm
L^∞	$\ x\ _\infty = \max_i x_i $	max element
Frobenius	$\ A\ _F = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\text{tr}(AA^\top)}$	matrix size

Quick Reference — A vs $A^\top$ vs A^{-1}

Notation	Name	What it does	How it looks
A	Original matrix	Transforms input	unchanged
$A^\top$	Transpose	Swaps rows and columns	$(A^\top)_{i,j} = A_{j,i}$
A^{-1}	Inverse	Undoes A — like dividing	$A^{-1}A = I$

Starting with the same matrix A :

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Transpose $A^\top$ — flip across diagonal (rows become columns):

$$A^\top = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} \rightarrow \text{row 1 } [1, 2] \text{ becomes col 1 } \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Inverse A^{-1} — the “undo” matrix, found by calculation (not just flipping!):

$$A^{-1} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix} \rightarrow A^{-1}A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Special case — Orthogonal matrix: transpose = inverse ($A^\top = A^{-1}$)

→ only works because orthogonal matrices purely rotate (no stretching).

Exam Trap: $A^\top$ is always easy — just flip.

A^{-1} requires real computation — unless A is orthogonal (then $A^{-1} = A^\top$ for free).

Part 4 — Identity, Inverse, Solving Systems

- Used for: solving systems of equations $Ax = b$; finding model parameters given data

$$A^{-1}A = I \quad Ax = b \Rightarrow x = A^{-1}b$$

Example — solve $Ax = b$:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} \quad b = \begin{pmatrix} 6 \\ 8 \end{pmatrix} \rightarrow A^{-1} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/4 \end{pmatrix} \rightarrow x = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

How to compute A^{-1} — 2×2 formula (exam):

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad \text{where } \det(A) = ad - bc$$

Steps: ① swap a and d on diagonal ② flip signs of b and c ③ divide by $\det(A)$

Example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \rightarrow \det(A) = 1 \cdot 4 - 2 \cdot 3 = -2 \rightarrow A^{-1} = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}$$

$$\text{Verify: } A^{-1}A = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \checkmark$$

Exam Trap: If $\det(A) = ad - bc = 0 \rightarrow$ division by zero $\rightarrow$ **no inverse exists** (singular matrix).

For larger matrices: use Gauss-Jordan — write $[A \mid I]$, row-reduce until left side $= I$, right side becomes A^{-1} .

A^{-1} exists only if: A is square AND columns are linearly independent (non-singular).

Situation	Solutions
$m > n$ (overdetermined)	None (generically)
$m < n$ (underdetermined)	Infinitely many
$m = n$, independent cols	Exactly one

Part 5 — Special Matrices

Diagonal:

- Used for: efficient scaling; eigenvalue matrices in decompositions; fast inversion

$$\text{diag}(v)x = v \odot x \quad \text{diag}(v)^{-1} = \text{diag}([1/v_1, \dots, 1/v_n]^T)$$

$$\text{diag}\left(\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}\right)x = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2x_1 \\ 3x_2 \\ 4x_3 \end{pmatrix} \rightarrow \text{same as } \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \odot x$$

Symmetric:

- Used for: covariance matrices, distance matrices, any matrix of pairwise values

$$A = A^T$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 4 \\ 3 & 4 & 6 \end{pmatrix}^{\top} \rightarrow \text{mirrored across diagonal}$$

Orthogonal (rows AND cols are orthonormal):

- *Used for:* rotation matrices, V and U in SVD, efficient inversion (just transpose)

Exam Trap: Symmetric $\neq$ Orthogonal — completely different properties!

Symmetric: $A = A^{\top}$ (mirrored across diagonal).

Orthogonal: $A^{\top}A = I$ (columns perpendicular + unit length). A matrix can be orthogonal without looking symmetric at all.

How to check if a matrix is orthogonal — verify two conditions on every column:

1. **Unit length:** $\|c_i\|_2 = 1$ — each column has length 1
2. **Perpendicular:** $c_i^{\top}c_j = 0$ for all $i \neq j$ — every pair of columns is at 90°

If both hold $\rightarrow$ orthogonal $\rightarrow$ use $A^{-1} = A^{\top}$ (free inverse, no computation).

Example — check col 1 and col 2 of $A = \frac{1}{3} \begin{pmatrix} 2 & -2 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{pmatrix}$:

First, extract the columns (each column = one vertical slice of the matrix):

$$c_1 = \frac{1}{3} \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \quad c_2 = \frac{1}{3} \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} \quad c_3 = \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$

Check unit length — square each element, sum them, take the square root. Must equal 1:

$$\|c_1\|_2 = \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2} = \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{4}{9}} = \sqrt{\frac{9}{9}} = \sqrt{1} = 1 \checkmark$$

Check perpendicular — dot product = multiply matching elements, then sum. Must equal 0:

$$\begin{aligned} c_1^{\top}c_2 &= \frac{2}{3} \cdot \frac{-2}{3} + \frac{1}{3} \cdot \frac{2}{3} + \frac{2}{3} \cdot \frac{1}{3} \\ &= \frac{-4}{9} + \frac{2}{9} + \frac{2}{9} = \frac{-4+2+2}{9} = \frac{0}{9} = 0 \checkmark \end{aligned}$$

Same check for every other pair: $c_1^{\top}c_3 = 0$ and $c_2^{\top}c_3 = 0$ (verify yourself!).

$$A^{\top}A = AA^{\top} = I \Rightarrow A^{-1} = A^{\top}$$

Inverting an orthogonal matrix = just transposing it — no computation needed.

Part 6 — Eigendecomposition

- *Used for:* understanding what a matrix does geometrically; PCA; finding principal directions of data; quadratic optimisation

$$Av = \lambda v$$

- v = eigenvector (direction unchanged by A , unit length $\|v\| = 1$)

- λ = eigenvalue (scaling factor along that direction)

How to compute eigenvalues — solve the characteristic equation $\det(A - \lambda I) = 0$:

Step 1: Subtract λ from every diagonal element $\rightarrow$ get $A - \lambda I$

Step 2: Compute $\det(A - \lambda I) = 0 \rightarrow$ gives a polynomial in λ

Step 3: Solve for λ using the quadratic formula (for 2×2):

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example — find eigenvalues of $B = \begin{pmatrix} 3 & 12 \\ 12 & 56 \end{pmatrix}$:

$$\text{Step 1: } B - \lambda I = \begin{pmatrix} 3 - \lambda & 12 \\ 12 & 56 - \lambda \end{pmatrix}$$

$$\text{Step 2: } \det = (3 - \lambda)(56 - \lambda) - 12 \cdot 12 = \lambda^2 - 59\lambda + 24 = 0$$

$$\text{Step 3: } \lambda = \frac{59 \pm \sqrt{59^2 - 4 \cdot 24}}{2} = \frac{59 \pm \sqrt{3385}}{2} \rightarrow \lambda_1 \approx 58.59, \quad \lambda_2 \approx 0.41$$

Example — $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ has eigenvalues $\lambda_1 = 4, \lambda_2 = 2$:

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rightarrow \text{direction } [1, 1]^\top \text{ scaled by } 4$$

$$A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} \rightarrow \text{direction } [1, -1]^\top \text{ scaled by } 2$$

Full decomposition:

$$A = V \text{diag}(\lambda) V^{-1} \quad V = [\mathbf{v}^{(1)}, \dots, \mathbf{v}^{(n)}]$$

Intuition: **rotate** (by V) $\rightarrow$ **scale** (by λ) $\rightarrow$ **rotate back** (V^{-1}).

For real symmetric matrices: n real eigenvalues, orthogonal eigenvectors $\rightarrow V^{-1} = V^\top$.

Optimization (exam!):

$$\max_{\|\mathbf{x}\|=1} \mathbf{x}^\top A \mathbf{x} \rightarrow \mathbf{x} = \text{eigenvector of max eigenvalue}$$

$$\min_{\|\mathbf{x}\|=1} \mathbf{x}^\top A \mathbf{x} \rightarrow \mathbf{x} = \text{eigenvector of min eigenvalue}$$

Part 7 — Singular Value Decomposition (SVD)

- *Used for:* pseudoinverse, data compression, dimensionality reduction (PCA), recommender systems, solving any linear system

Works for **any** matrix (generalization of eigendecomposition):

$$A = U D V^\top$$

- U : orthogonal — left singular vectors (output directions)

- D : diagonal — singular values (how much each direction is stretched)
- V : orthogonal — right singular vectors (input directions)

Full worked example — SVD of $A = \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$ (shape 3×2):

Step 1 — Compute $A^\top A$, find eigenvalues (used for both D and V):

$$A^\top A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 9 & 0 \\ 0 & 4 \end{pmatrix} \rightarrow \lambda_1 = 9, \quad \lambda_2 = 4$$

Step 2 — Build D (square roots, same shape as A):

$$D = \begin{pmatrix} \sqrt{9} & 0 \\ 0 & \sqrt{4} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$$

Step 3 — Build V (solve $(A^\top A - \lambda I)\mathbf{v} = 0$ for each λ , stack as columns):

$$\lambda_1 = 9: \begin{pmatrix} 0 & 0 \\ 0 & -5 \end{pmatrix} \mathbf{v} = 0 \rightarrow \mathbf{v}^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \lambda_2 = 4: \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix} \mathbf{v} = 0 \rightarrow \mathbf{v}^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$V = [\mathbf{v}^{(1)} \mid \mathbf{v}^{(2)}] = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Step 4 — Compute $AA^\top$, find eigenvectors (for U):

$$AA^\top = \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \lambda_1 = 9, \lambda_2 = 4, \lambda_3 = 0$$

$$\mathbf{u}^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \mathbf{u}^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \mathbf{u}^{(3)} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$U = [\mathbf{u}^{(1)} \mid \mathbf{u}^{(2)} \mid \mathbf{u}^{(3)}] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Verify: $UDV^\top = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} = A \checkmark$

Tip: Key insight: eigenvectors = directions. Stack them side by side as columns — col 1 = biggest λ , col 2 = next, and so on.

How to build U — eigenvectors of $AA^\top$, stacked as columns in descending λ order. Size: $m \times m$ (m = rows of A):

$$U = [\mathbf{u}^{(1)} \mid \mathbf{u}^{(2)} \mid \dots \mid \mathbf{u}^{(m)}] \quad \text{col } i = \text{eigenvector of } AA^\top \text{ for } \lambda_i$$

Example — A is $3 \times 2 \rightarrow AA^\top$ is $3 \times 3 \rightarrow U$ is 3×3 :

$$U = \begin{pmatrix} -0.283 & -0.868 & 0.408 \\ -0.538 & -0.208 & -0.816 \\ -0.794 & 0.451 & 0.408 \end{pmatrix}$$

How to build D — $\sqrt{\lambda_i}$ of $A^\top A$ on diagonal, descending order. Size = same shape as A :

$$D = \begin{pmatrix} \sqrt{\lambda_1} & 0 & \dots \\ 0 & \sqrt{\lambda_2} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix} \text{ same shape as } A, \text{ extra rows/cols} = 0$$

Example — A is 3×2 , $\lambda_1 = 58.59$, $\lambda_2 = 0.41$:

$$D = \begin{pmatrix} \sqrt{58.59} & 0 \\ 0 & \sqrt{0.41} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 7.65 & 0 \\ 0 & 0.64 \\ 0 & 0 \end{pmatrix}$$

How to build V — eigenvectors of $A^\top A$, stacked as columns in descending λ order. Size: $n \times n$ ($n = \text{cols of } A$):

$$V = [\mathbf{v}^{(1)} \mid \mathbf{v}^{(2)} \mid \dots \mid \mathbf{v}^{(n)}] \quad \text{col } i = \text{eigenvec of } A^\top A \text{ for } \lambda_i$$

Example — A has 2 cols $\rightarrow V$ is 2×2 :

$$V = \begin{pmatrix} -0.211 & -0.977 \\ -0.977 & 0.211 \end{pmatrix}$$

Tip: Full SVD recipe for $A = UDV^\top$:

1. U : eigenvectors of $AA^\top$ as columns ($m \times m$)
2. D : $\sqrt{\lambda_i}$ on diagonal, same shape as A ($m \times n$)
3. V : eigenvectors of $A^\top A$ as columns ($n \times n$)

All in descending eigenvalue order.

Part 8 — Moore-Penrose Pseudoinverse

- *Used for:* least-squares regression, solving overdetermined systems (more equations than unknowns); any time a true inverse doesn't exist

When A^{-1} doesn't exist:

$$A^+ = VD^+U^\top \quad (D^+ : \text{reciprocal of each non-zero diagonal of } D)$$

Example — $D = \begin{pmatrix} 3 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} \rightarrow D^+ = \begin{pmatrix} 1/3 & 0 & 0 \\ 0 & 1/2 & 0 \end{pmatrix}$

Case	Result
More rows than cols (overdetermined)	Minimises $\ A\mathbf{x} - \mathbf{b}\ _2$ (least squares)
More cols than rows (underdetermined)	Minimises $\ \mathbf{x}\ _2$ among all solutions

Part 9 — Trace

- *Used for:* computing Frobenius norm, simplifying matrix derivative expressions, checking eigenvalue sums

$$\text{tr}(A) = \sum_i A_{i,i} = \sum_i \lambda_i$$

Example:

$$\text{tr}\left(\begin{pmatrix} 1 & 9 & 7 \\ 2 & 5 & 3 \\ 4 & 6 & 8 \end{pmatrix}\right) = 1 + 5 + 8 = 14$$

- $\text{tr}(A^\top) = \text{tr}(A)$
- $\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA) \leftarrow$ cyclic invariant
- $\|A\|_F = \sqrt{\text{tr}(AA^\top)}$

Part 10 — Determinant

- *Used for:* checking if a matrix is invertible; measuring volume scaling; appears in probability density formulas (multivariate Gaussian)

$$\det(A) = \prod_i \lambda_i$$

Example:

$$\det\left(\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}\right) = 3 \cdot 3 - 1 \cdot 1 = 8 \rightarrow \lambda_1 \cdot \lambda_2 = 4 \cdot 2 = 8 \checkmark$$

- $\det(A) = 0 \rightarrow$ singular, not invertible (at least one eigenvalue is zero)
- $\det(A) \neq 0 \rightarrow$ invertible
- Geometric meaning: factor by which A scales volume

Cheat Sheet**WEEK 1 — LINEAR ALGEBRA CHEAT SHEET**

Structures: scalar=0D · vector=1D(col) · matrix=2D · tensor=3D+

Multiply:

- Standard: $C_{i,j} = \sum_k A_{i,k} B_{k,j}$
- Dot product: $\mathbf{x}^\top \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta$
- Hadamard $\odot$: $C_{i,j} = A_{i,j} B_{i,j}$
- $AB \neq BA$ — NEVER commutative

Norms: $L^2 = \sqrt{\sum x^2}$ · $L^1 = \sum |x|$ · $L^\infty = \max |x|$ · $\|A\|_F = \sqrt{\text{tr}(AA^\top)}$

Special: Symmetric $A = A^\top$ · Orthogonal $A^{-1} = A^\top$ · Diagonal: invert = reciprocal diagonal

Eigen: $A\mathbf{v} = \lambda\mathbf{v} \rightarrow A = V \text{diag}(\lambda) V^{-1}$

SVD: $A = U D V^\top$ (always works)

Pseudoinverse: $A^+ = V D^+ U^\top$

Trace: $\text{tr}(A) = \sum A_{i,i} = \sum \lambda_i$ (cyclic invariant)

Det: $\det(A) = \prod \lambda_i$ · zero = singular = not invertible

Week 2: Probability and Information Theory

Chapter 2: Probability and Information Theory

Part 1 — Random Variables

- *Used for:* describing uncertain quantities; modelling data distributions; defining loss functions

A **random variable** x can take on different values with associated probabilities.

Type	Values	Described by	Notation
Discrete	Countable set	Probability Mass Function (PMF)	$P(x = k)$
Continuous	Real interval	Probability Density Function (PDF)	$p(x)$

PMF (discrete) — probability at each exact value:

$$P(x = k) \geq 0 \quad \sum_k P(x = k) = 1$$

PDF (continuous) — probability density; integrate over an interval to get probability:

$$p(x) \geq 0 \quad \int_{-\infty}^{\infty} p(x) dx = 1 \quad P(a \leq x \leq b) = \int_a^b p(x) dx$$

Exam Trap: $p(x)$ is a **density**, not a probability. $p(x) > 1$ is valid! Only the integral over an interval gives probability.

Part 2 — Key Distributions

Bernoulli — single binary outcome ($x \in \{0, 1\}$):

- *Used for:* modelling coin flips, binary classification outputs, gates in LSTMs

$$P(x = 1) = \varphi, \quad P(x = 0) = 1 - \varphi$$

$$\mathbb{E}[x] = \varphi \quad \text{Var}(x) = \varphi(1 - \varphi)$$

Example: $\varphi = \frac{3}{4} \rightarrow \mathbb{E}[x] = \frac{3}{4}, \text{Var}(x) = \frac{3}{16}$.

Uniform (continuous) — equal probability over interval $[a, b]$:

- *Used for:* initialising weights, random sampling, baseline distributions

$$p(x) = \frac{1}{b - a} \quad x \in [a, b], \quad \text{else } 0$$

$$\mathbb{E}[x] = \frac{a + b}{2} \quad \text{Var}(x) = \frac{(b - a)^2}{12}$$

Example: $x \in [-0.5, 1] \rightarrow p(x) = \frac{2}{3}, \mathbb{E}[x] = \frac{1}{4}, \text{Var}(x) = \frac{3}{16}$.

Gaussian (Normal) — bell curve, most important distribution in ML:

- *Used for:* modelling real-valued data, noise, weight initialisation, latent spaces in VAEs

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$\mathbb{E}[x] = \mu \quad \text{Var}(x) = \sigma^2$$

Standard Normal: $\mu = 0, \sigma = 1$. Notation: $x \sim \mathcal{N}(\mu, \sigma^2)$.

Tip: Why Gaussian? The Central Limit Theorem: sums of many independent random variables converge to a Gaussian. Most real-world noise is approximately Gaussian.

Exponential — models time between events; heavy tail to the right:

$$p(x) = \lambda \exp(-\lambda x), \quad x \geq 0 \quad \mathbb{E}[x] = \frac{1}{\lambda}$$

Laplace — like Gaussian but with heavier tails; used in L1 regularisation:

$$p(x) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right)$$

Part 3 — Expectations and Variance

- *Used for:* computing loss function values, quantifying spread, deriving gradient updates

Discrete: $\mathbb{E}[f(x)] = \sum_x f(x)P(x)$

Continuous: $\mathbb{E}[f(x)] = \int f(x)p(x)dx$

Variance: $\text{Var}(x) = \mathbb{E}[x^2] - (\mathbb{E}[x])^2 = \mathbb{E}[(x - \mathbb{E}[x])^2]$

Key properties:

- $\mathbb{E}[ax + b] = a, \mathbb{E}[x] + b$ — linearity
- $\text{Var}(ax) = a^2 \text{Var}(x)$
- $\text{Var}(x + y) = \text{Var}(x) + \text{Var}(y)$ only if x, y are independent

Covariance — measures linear dependence between two variables:

$$\text{Cov}(x, y) = \mathbb{E}[(x - \mathbb{E}[x])(y - \mathbb{E}[y])] = \mathbb{E}[xy] - \mathbb{E}[x]\mathbb{E}[y]$$

Part 4 — Bayes' Theorem

- *Used for:* updating beliefs given evidence; Naive Bayes classifiers; probabilistic inference in all Bayesian models

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

- $P(A)$ — **prior**: belief before seeing evidence

- $P(B | A)$ — **likelihood**: probability of evidence given hypothesis
- $P(A | B)$ — **posterior**: updated belief after evidence
- $P(B)$ — **marginal** (normalisation constant)

Law of total probability (for computing $P(B)$):

$$P(B) = \sum_i P(B | A_i)P(A_i)$$

Example — cab problem: 5000 yellow cabs, 100 red cabs; witness accuracy 95%. Find $P(c = r | w = r)$:

$$P(c = r) = \frac{100}{5100} \approx 0.0196$$

$$P(w = r) = 0.95 \cdot \frac{100}{5100} + 0.05 \cdot \frac{5000}{5100} \approx 0.0676$$

$$P(c = r | w = r) = \frac{0.0196 \cdot 0.95}{0.0676} \approx 27.5\%$$

Exam Trap: Base rate neglect: even with a 95%-accurate witness, the answer is only 27.5%. Red cabs are so rare that most “red” reports are still wrong yellow cabs. Always account for the prior!

Bayesian classification — assign x to the class with highest posterior:

$$\text{assign to } c_i \quad \text{if} \quad P(y = c_i)p(x | y = c_i) > P(y = c_j)p(x | y = c_j)$$

The denominator $p(x)$ is the same for all classes and cancels.

Part 5 — Information Theory

- *Used for:* cross-entropy loss (classifier training), measuring model uncertainty, defining KL divergence for VAEs

Self-information — surprise of a single outcome:

$$I(x) = -\log P(x)$$

Rare events $\rightarrow$ large $I(x)$ (more surprising). Common events $\rightarrow$ small $I(x)$.

Log base: $\log_2 = \text{bits}$; $\ln = \text{nats}$. Deep learning typically uses nats.

Shannon Entropy — expected surprise (average information content):

Discrete: $H(x) = -\sum_x P(x) \log P(x) = \mathbb{E}[-\log P(x)]$

Gaussian: $H(x) = \frac{1}{2}(\log(2\pi\sigma^2) + 1)$

Higher variance $\rightarrow$ higher entropy. Uniform distribution has maximum entropy.

Example:

- $\sigma_1^2 = 0.26 \rightarrow H(x | y = c_1) = 0.5(\ln(2\pi \cdot 0.26) + 1) \approx 0.745 \text{ nats}$
- $\sigma_2^2 = 0.99 \rightarrow H(x | y = c_2) = 0.5(\ln(2\pi \cdot 0.99) + 1) \approx 1.414 \text{ nats}$

Cross-Entropy — entropy of distribution P measured under model Q :

$$H(P, Q) = - \sum_x P(x) \log Q(x) = H(P) + D_{\text{KL}}(P \parallel Q)$$

Used as the standard loss for classification (comparing true labels P to model predictions Q).

KL Divergence — measures how much Q differs from P :

$$D_{\text{KL}}(P \parallel Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)} \geq 0$$

Exam Trap: $D_{\text{KL}}(P \parallel Q) \neq D_{\text{KL}}(Q \parallel P)$ — KL divergence is **asymmetric**. It is not a distance metric!

- $D_{\text{KL}}(P \parallel Q) = 0$ iff $P = Q$ everywhere.
- Minimising $H(P, Q)$ over Q is equivalent to minimising $D_{\text{KL}}(P \parallel Q)$ (since $H(P)$ is constant).

Tip: Cross-entropy loss in practice:

$$\mathcal{L} = - \sum_k y_k \log \hat{y}_k$$

where y_k = true label (one-hot), $\hat{y}_k$ = softmax output. This is exactly $H(P, Q)$.

Cheat Sheet

WEEK 2 — PROBABILITY & INFORMATION THEORY CHEAT SHEET

Random Variables:

- Discrete $\rightarrow$ PMF: $\sum P(x) = 1$
- Continuous $\rightarrow$ PDF: $\int p(x)dx = 1$; $p(x)$ can exceed 1!

Key Distributions:

- Bernoulli(φ): $\mathbb{E} = \varphi$, $\text{Var} = \varphi(1 - \varphi)$
- Uniform[a, b]: $p = \frac{1}{b-a}$, $\mathbb{E} = \frac{a+b}{2}$, $\text{Var} = \frac{(b-a)^2}{12}$
- Gaussian(μ, σ^2): $p = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

Expectation & Variance:

- $\mathbb{E}[f(x)] = \sum f(x)P(x)$ or $\int f(x)p(x)dx$
- $\text{Var}(x) = \mathbb{E}[x^2] - (\mathbb{E}[x])^2$

Bayes: $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$ — always account for the prior!

Classification: assign to c_i if $P(c_i)p(x|c_i)$ is largest.

Entropy:

- Discrete: $H = - \sum P \log P$
- Gaussian: $H = \frac{1}{2}(\log(2\pi\sigma^2) + 1)$

Cross-entropy: $H(P, Q) = - \sum P \log Q = H(P) + D_{\text{KL}}(P \parallel Q)$

KL-Divergence: $D_{\text{KL}}(P\|Q) = \sum P \log \frac{P}{Q} \geq 0$; asymmetric!